

Experiment 2 - DDL Constraints

Question 1

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
-- 1
ALTER TABLE STUDENT2
ADD PRIMARY KEY(RegNo);

-- 2(a)
ALTER TABLE FACULTY
ADD PRIMARY KEY(FacNo);

-- 2(b)
ALTER TABLE FACULTY
ADD CONSTRAINT chk_gender
```

The Result Grid shows the output of the first query, displaying the columns: RegNo, SName, Gender, DOB, MobileNo, and City. The data is as follows:

RegNo	SName	Gender	DOB	MobileNo	City
101	Arun	M	2004-01-15	9876543210	Chennai
102	Pritha	F	2003-05-12	9876543211	Madurai
103	Rahul	M	2004-07-20	9876543212	Coimbatore
104	Divya	F	2003-11-25	9876543213	Salem
105	Karthik	M	2004-03-10	9876543214	Trichy

The Output pane shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
11	21:10:53	CREATE TABLE COURSE (CourseNo INT, CourseDesc VARCHAR(40), CourseType CHAR(1), Se...	0 row(s) affected	1.110 sec
12	21:10:58	INSERT INTO COURSE VALUES (501,'Database Management','T','3','101','1',502,'Operating System','T','4'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.234 sec
13	21:11:02	CREATE TABLE STUDENT3 (RegNo INT, Attendance INT)	0 row(s) affected	0.672 sec
14	21:11:07	INSERT INTO STUDENT3 VALUES (101,90), (102,88), (103,95), (104,92), (105,85)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.047 sec
15	21:11:14	ALTER TABLE STUDENT2 ADD PRIMARY KEY(RegNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.469 sec
16	21:11:21	SELECT * FROM STUDENT2 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Figure 1: Question 1 - Add PRIMARY KEY to STUDENT2

Question 2(a)

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
-- 1
ALTER TABLE STUDENT2
ADD PRIMARY KEY(RegNo);

-- 2(a)
ALTER TABLE FACULTY
ADD PRIMARY KEY(FacNo);

-- 2(b)
ALTER TABLE FACULTY
ADD CONSTRAINT chk_gender
```

The Result Grid shows the output of the first query, displaying the columns: FacNo, FacName, Gender, DOB, DOJ, and MobileNo. The data is as follows:

FacNo	FacName	Gender	DOB	DOJ	MobileNo
1	Ramesh	M	1985-06-15	2012-07-01	9876500001
2	Lalshmi	F	1987-09-10	2014-06-15	9876500002
3	Suresh	M	1982-12-20	2010-01-12	9876500003
4	Anitha	F	1990-03-18	2016-05-20	9876500004

The Output pane shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
13	21:11:02	CREATE TABLE STUDENT3 (RegNo INT, Attendance INT)	0 row(s) affected	0.672 sec
14	21:11:07	INSERT INTO STUDENT3 VALUES (101,90), (102,88), (103,95), (104,92), (105,85)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.047 sec
15	21:11:14	ALTER TABLE STUDENT2 ADD PRIMARY KEY(RegNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.469 sec
16	21:11:21	SELECT * FROM STUDENT2 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
17	21:12:10	ALTER TABLE FACULTY ADD PRIMARY KEY(FacNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.563 sec
18	21:12:18	SELECT * FROM FACULTY LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Figure 2: Question 2(a) - Add PRIMARY KEY to FACULTY

Question 2(b) - Add CHECK Constraint to FACULTY

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
-- 2(b)
109
110 ALTER TABLE FACULTY
111 ADD CONSTRAINT chk_gender
112 CHECK(Gender IN ('M','F'));
113
-- 3
114
115 ALTER TABLE DEPARTMENT
116 ADD PRIMARY KEY(DepNo);
117
-- 4(a)
118
119 ALTER TABLE COURSE
120 ADD PRIMARY KEY(CourseNo);
```

The Result Grid shows the output of the queries:

#	Time	Action	Message	Duration / Patch
15	21:11:14	ALTER TABLE STUDENT2 ADD PRIMARY KEY(RegNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.469 sec
16	21:11:21	SELECT * FROM STUDENT2 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
17	21:12:10	ALTER TABLE FACULTY ADD PRIMARY KEY(FacNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.563 sec
18	21:12:18	SELECT * FROM FACULTY LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
19	21:12:54	ALTER TABLE FACULTY ADD CONSTRAINT chk_gender CHECK(Gender IN ('M','F'))	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	1.516 sec
20	21:13:00	SELECT * FROM FACULTY LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

The Output pane shows the Action Output for the queries.

Figure 3: Question 2(b) - Add CHECK Constraint to FACULTY

Question 3 - Add PRIMARY KEY to DEPARTMENT

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
112 CHECK(Gender IN ('M','F'));
113
-- 3
114
115 ALTER TABLE DEPARTMENT
116 ADD PRIMARY KEY(DepNo);
117
-- 4(a)
118
119 ALTER TABLE COURSE
120 ADD PRIMARY KEY(CourseNo);
121
-- 4(b)
122
123 ALTER TABLE COURSE
```

The Result Grid shows the output of the queries:

#	Time	Action	Message	Duration / Patch
17	21:12:10	ALTER TABLE FACULTY ADD PRIMARY KEY(FacNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.563 sec
18	21:12:18	SELECT * FROM FACULTY LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
19	21:12:54	ALTER TABLE FACULTY ADD CONSTRAINT chk_gender CHECK(Gender IN ('M','F'))	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	1.516 sec
20	21:13:00	SELECT * FROM FACULTY LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
21	21:13:19	ALTER TABLE DEPARTMENT ADD PRIMARY KEY(DepNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.203 sec
22	21:13:24	SELECT * FROM DEPARTMENT LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

The Output pane shows the Action Output for the queries.

Figure 4: Question 3 - Add PRIMARY KEY to DEPARTMENT

Question 4(a) - Add PRIMARY KEY to COURSE

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

```
-- 4(a)
ALTER TABLE COURSE
ADD PRIMARY KEY(CourseNo);
```

The Result Grid displays the following data:

CourseNo	CourseDesc	CourseType	SemNo	HallNo	FacNo
501	Database Management	T	3	H101	1
502	Operating Systems	T	4	H102	2
503	Computer Networks	T	5	H103	3
504	Data Structures	P	2	L201	1
505	Java Programming	T	6	H104	4

The Output pane shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
19	21:12:54	ALTER TABLE FACULTY ADD CONSTRAINT chk_gender CHECK(Gender IN ('M','F'))	4 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.516 sec
20	21:13:00	SELECT * FROM FACULTY LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
21	21:13:19	ALTER TABLE DEPARTMENT ADD PRIMARY KEY(DepNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.203 sec
22	21:13:24	SELECT * FROM DEPARTMENT LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
23	21:13:38	ALTER TABLE COURSE ADD PRIMARY KEY(CourseNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.140 sec
24	21:13:46	SELECT * FROM COURSE LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Figure 5: Question 4(a) - Add PRIMARY KEY to COURSE

Question 4(b) - Add CHECK Constraint to COURSE

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

```
-- 4(b)
ALTER TABLE COURSE
ADD CONSTRAINT chk_sem
CHECK(SemNo BETWEEN 1 AND 6);
```

The Result Grid displays the following data:

CourseNo	CourseDesc	CourseType	SemNo	HallNo	FacNo
501	Database Management	T	3	H101	1
502	Operating Systems	T	4	H102	2
503	Computer Networks	T	5	H103	3
504	Data Structures	P	2	L201	1
505	Java Programming	T	6	H104	4

The Output pane shows the execution results of the queries:

#	Time	Action	Message	Duration / Fetch
21	21:13:19	ALTER TABLE DEPARTMENT ADD PRIMARY KEY(DepNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.203 sec
22	21:13:24	SELECT * FROM DEPARTMENT LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
23	21:13:38	ALTER TABLE COURSE ADD PRIMARY KEY(CourseNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.140 sec
24	21:13:46	SELECT * FROM COURSE LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
25	21:13:59	ALTER TABLE COURSE ADD CONSTRAINT chk_sem CHECK(SemNo BETWEEN 1 AND 6)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	1.516 sec
26	21:14:20	SELECT * FROM COURSE LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec

Figure 6: Question 4(b) - Add CHECK Constraint to COURSE

Question 5 - Add UNIQUE Constraint to STUDENT2

The screenshot displays the MySQL Workbench interface. The 'Query Editor' window shows the following SQL script:

```
125 CHECK(SenNo BETWEEN 1 AND 6) ;
126
127 -- 5
128 * ALTER TABLE STUDENT2
129 ADD CONSTRAINT uk_student_name
130 UNIQUE(S1Name) ;
131
132 -- 6
133 * ALTER TABLE STUDENT3
134 ADD CONSTRAINT fk_reg
135 FOREIGN KEY(RegNo)
136 REFERENCES STUDENT2(RegNo) ;
```

The 'Result Grid' shows the execution results of the queries. The first query, 'ALTER TABLE STUDENT2 ADD CONSTRAINT uk_student_name UNIQUE(S1Name) ;', was executed successfully at 21:14:52. The output shows 0 rows affected, 0 duplicates, and 0 warnings. The duration was 0.485 seconds.

#	Time	Action	Message	Duration / Fetch
23	21:13:38	ALTER TABLE COURSE ADD PRIMARY KEY(CourseNo)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	1.140 sec
24	21:13:46	SELECT * FROM COURSE LIMIT 0,1000	5 row(s) returned	0.000 sec / 0.000 sec
25	21:13:59	ALTER TABLE COURSE ADD CONSTRAINT chk_sen CHECK(SenNo BETWEEN 1 AND 6)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	1.516 sec
26	21:14:20	SELECT * FROM COURSE LIMIT 0,1000	5 row(s) returned	0.016 sec / 0.000 sec
27	21:14:52	ALTER TABLE STUDENT2 ADD CONSTRAINT uk_student_name UNIQUE(S1Name)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.485 sec
28	21:14:57	SELECT * FROM STUDENT2 LIMIT 0,1000	5 row(s) returned	0.000 sec / 0.000 sec

Figure 7: Question 5 - Add UNIQUE Constraint to STUDENT2

Question 6 - Add FOREIGN KEY to STUDENT3

The screenshot displays the MySQL Workbench interface. The left sidebar shows a tree view of databases, including 'college', 'college_db', 'companydb', 'customer', 'dms', 'mag', 'saveetha', 'studentdb', 'sys', and 'xyz'. The main editor window shows a SQL script with the following queries:

```
125 CHECK(SemNo BETWEEN 1 AND 6) ;
126
127 -- 5
128 ALTER TABLE STUDENT2
129 ADD CONSTRAINT uk_student_name
130 UNIQUE(S1Name) ;
131
132 -- 6
133 ALTER TABLE STUDENT3
134 ADD CONSTRAINT fk_reg
135 FOREIGN KEY(RegNo)
136 REFERENCES STUDENT2(RegNo) ;
```

Below the SQL editor, the 'Result Grid' shows the output of the queries. The first query returns a single row with 'RegNo' 101 and 'Attendance' 90. The subsequent queries return no rows.

RegNo	Attendance
101	90
102	88
103	95
104	92
105	85

The 'Output' tab at the bottom shows the execution results of the queries. The first query, 'ALTER TABLE COURSE ADD CONSTRAINT ch_sem CHECK(SemNo BETWEEN 1 AND 6)', executed at 21:13:59 and affected 5 rows. The second query, 'SELECT * FROM COURSE LIMIT 0, 1000', executed at 21:14:20 and returned 5 rows. The third query, 'ALTER TABLE STUDENT2 ADD CONSTRAINT uk_student_name UNIQUE(S1Name)', executed at 21:14:52 and affected 0 rows. The fourth query, 'SELECT * FROM STUDENT2 LIMIT 0, 1000', executed at 21:14:57 and returned 5 rows. The fifth query, 'ALTER TABLE STUDENT3 ADD CONSTRAINT fk_reg FOREIGN KEY(RegNo) REFERENCES STUDENT2(RegNo)', executed at 21:15:09 and affected 5 rows. The final query, 'SELECT * FROM STUDENT3 LIMIT 0, 1000', executed at 21:15:15 and returned 5 rows.

#	Time	Action	Message	Duration / Fetch
25	21:13:59	ALTER TABLE COURSE ADD CONSTRAINT ch_sem CHECK(SemNo BETWEEN 1 AND 6)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	1.516 sec
26	21:14:20	SELECT * FROM COURSE LIMIT 0, 1000	5 row(s) returned	0.016 sec / 0.000 sec
27	21:14:52	ALTER TABLE STUDENT2 ADD CONSTRAINT uk_student_name UNIQUE(S1Name)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.485 sec
28	21:14:57	SELECT * FROM STUDENT2 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
29	21:15:09	ALTER TABLE STUDENT3 ADD CONSTRAINT fk_reg FOREIGN KEY(RegNo) REFERENCES STUDENT2(RegNo)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	1.281 sec
30	21:15:15	SELECT * FROM STUDENT3 LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Figure 8: Question 6 - Add FOREIGN KEY to STUDENT3